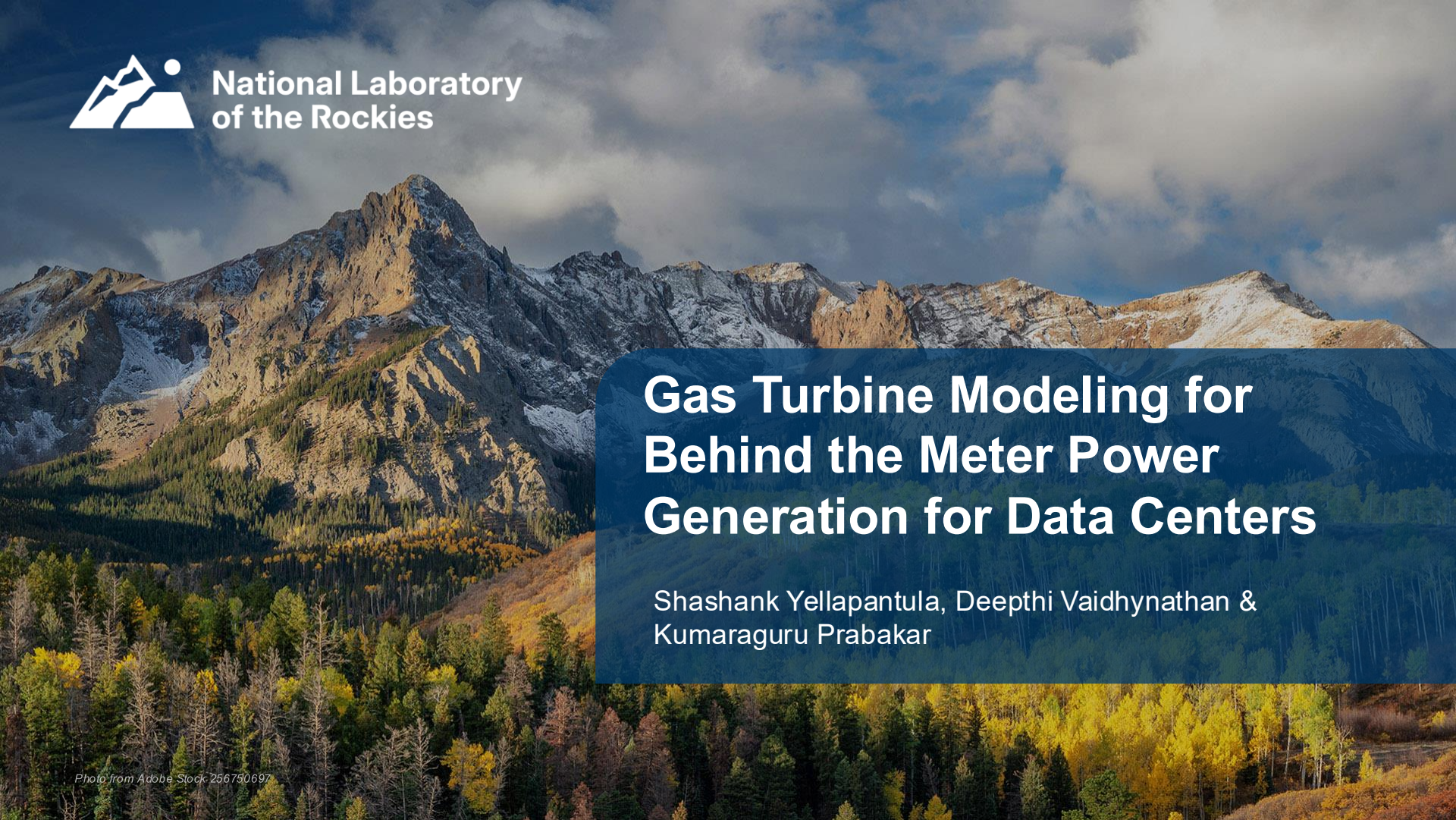


Gas Turbine Modeling for Behind the Meter Power Generation for Data Centers

Shashank Yellapantula, Deepthi Vaidhynathan & Kumaraguru Prabakar

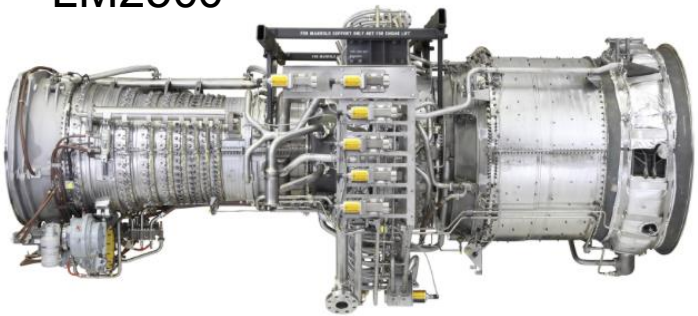
Photo from Adobe Stock 256750697

Aeroderivative Model

Developed a flexible framework in python to study various gas turbine cycles

- Model both simple and combined cycles with a heat-recovery steam cycle
- Can also be used as a dynamic dispatch model for a fleet of aeroderivatives

LM2500



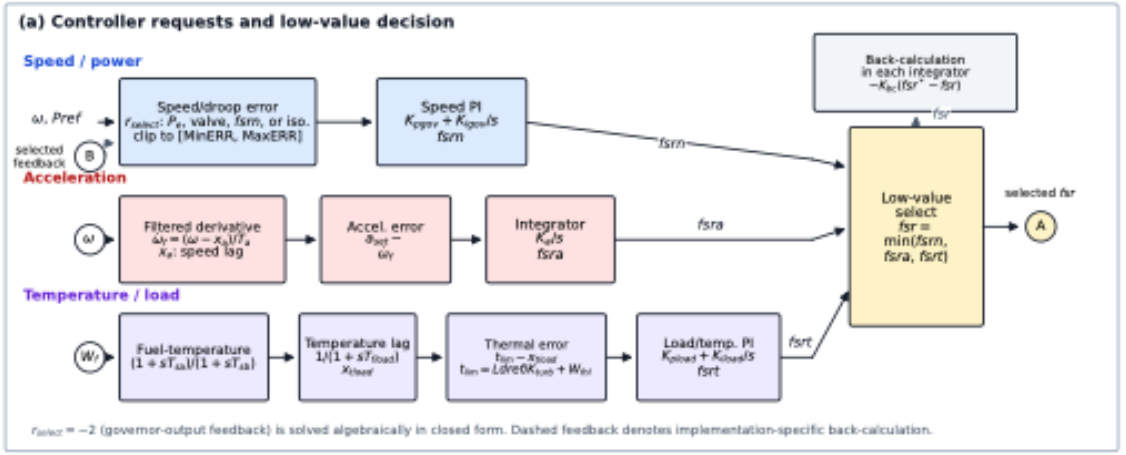
LM2500 modeled as multi-shaft system with

1. HP gas-generator rotor: the high-speed compressor/turbine core
2. PT/generator rotor: the power turbine and electrical generator, tied to 60 Hz power at 3600 rpm.

GGOV Governor Model

Governor model

- Implementation based on the IEEE PES GGOV model
- Include three competing controller with low-value select
  1. Speed/power governor
  2. Acceleration limiter
  3. Temperature limiter

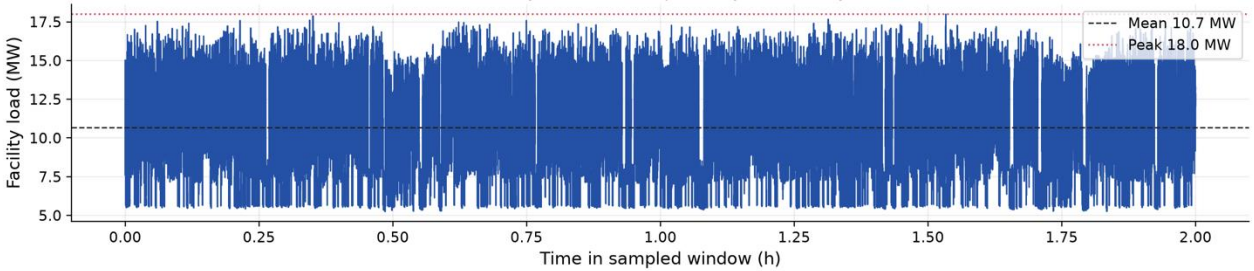


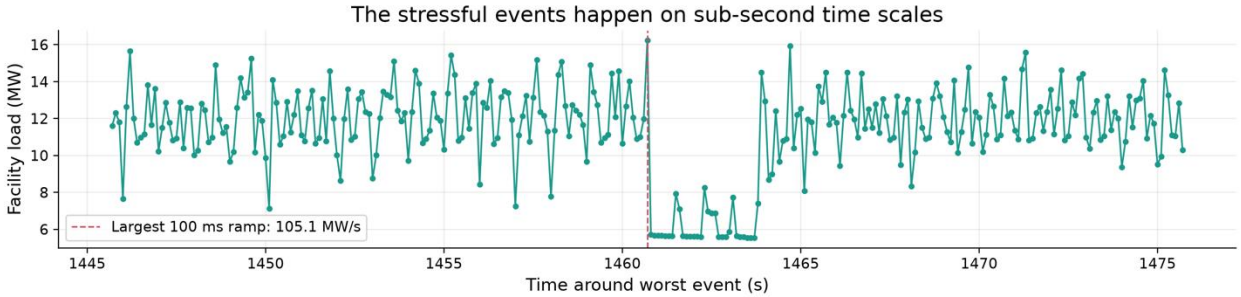
Developed a unique capability allowing us to study various power generation & storage options to simulate the power demand from various data centers

Data Center Load: Islanding Example

Data Center Load Profile

- Per GPU telemetry recorded every 100ms
- Used a 2h, 35 active GPU trace
- Scaled it to match power demand of 18 MW
- Worst case scenario of electrical load from a data center

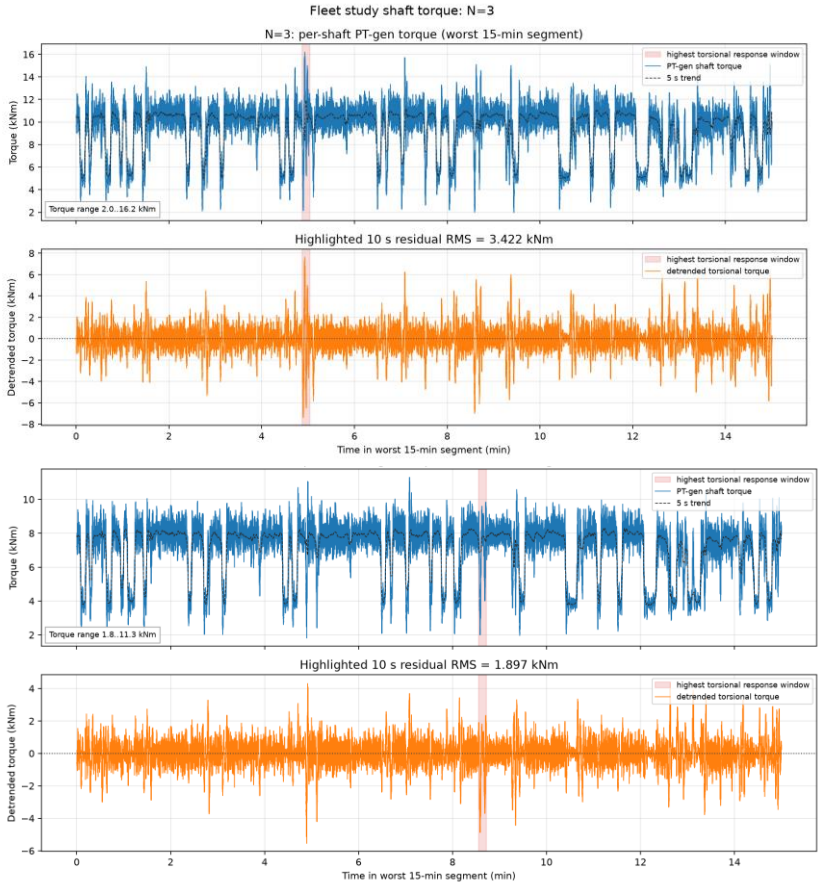




- Fleet with one (N=1) and two (N=2) LM2500s trip around these stressful events
- N = 3 and N = 4 ride through these events

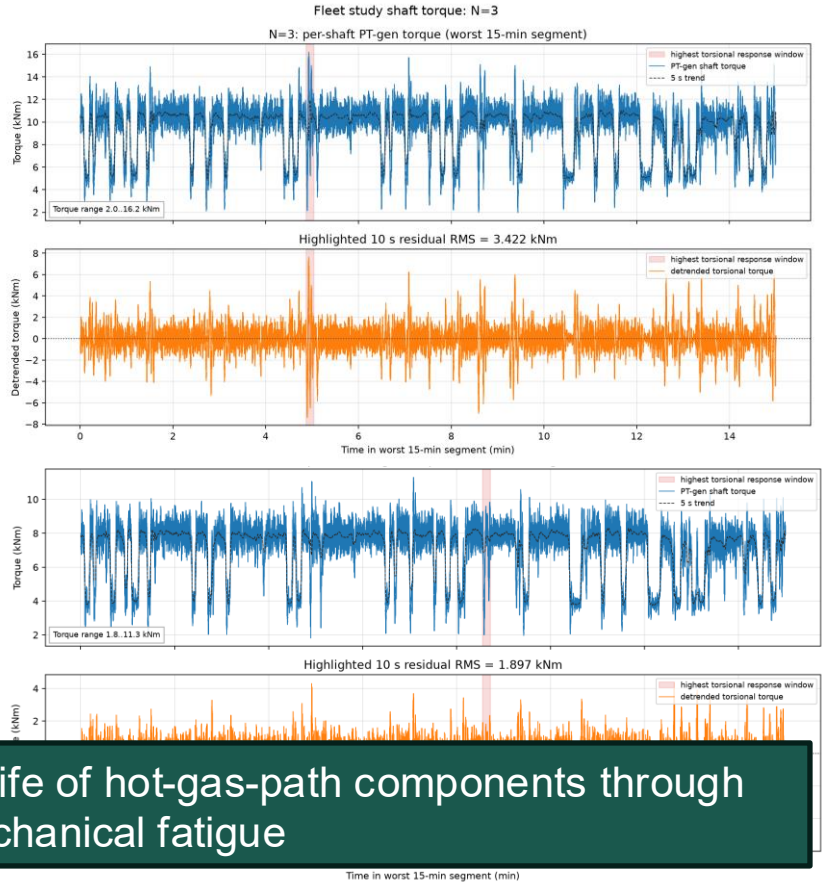
Fleet Modeling – Shaft Life

- Shaft Torque can be used to investigate
    1. High Cycle Fatigue (HCF): fast ringing of the PT-generator shaft
    2. Low Cycle Fatigue (LCF): slow torque swings caused by turbine governor
- Need exact shaft design and material properties from vendor to calculate impact on life.
- Between N=3/4 we observe
    1. D_{HCF} damage reduction N=3 -> N=4: 49.9x
    2. D_{LCF} damage reduction N=3 -> N=4: 3.7x



Fleet Modeling – Shaft Life

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    1. D_{HCF} damage reduction N=3 -> N=4: 49.9x



Repeated heating and cooling can shorten the life of hot-gas-path components through thermal and thermomechanical fatigue

Thank You

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